

On the hydrodynamic attractor of Yang-Mills plasma

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There is mounting evidence suggesting that relativistic hydrodynamics becomes relevant for the physics of quark-gluon plasma as the result of nonhydrodynamic modes decaying to an attractor apparent even when the system is far from local equilibrium. Here we determine this attractor for Bjorken flow in $\mathcal{N} = 4$ supersymmetric Yang-Mills theory (SYM) using Borel summation of the gradient expansion of the expectation value of the energy momentum tensor. By comparing the result to numerical simulations of the flow based on the AdS/CFT correspondence we show that it provides an accurate and unambiguous approximation of the hydrodynamic attractor in this system. This development has important implications for the formulation of effective theories of hydrodynamics.

Introduction. – Heavy-ion collision experiments and their phenomenological description have lead to the realization that relativistic hydrodynamics works very well rather far outside its traditionally understood domain of validity. Variants of Müller-Israel-Stewart theory [1, 2] have successfully been applied in rather extreme conditions, which could hardly be assumed to be close to local equilibrium. Furthermore, model calculations exist where it is possible to study the emergence of hydrodynamic behaviour and test to what extent an effective description in terms of hydrodynamics can match microscopic results [3].

Such calculations were originally carried out in $\mathcal{N} = 4$ SYM using the AdS/CFT correspondence [4, 5], but similar studies have since also been performed using kinetic theory [6, 7]. The conclusion from these investigations is that the domain of validity of a hydrodynamic description is delimited by the decay of nonhydrodynamic modes [4, 5, 8, 9]. The outcome of this transition to hydrodynamics (“hydrodynamization”) is that the system reaches a hydrodynamic attractor [10] which governs its subsequent evolution toward equilibrium. This attractor is a special solution to which generic histories decay exponentially in time well before local equilibrium sets in (see Fig. 1). It incorporates all orders of the hydrodynamic gradient expansion, and at sufficiently late times coincides with the predictions of relativistic Navier-Stokes theory. The existence of an attractor in this sense is a critically important issue for hydrodynamics, because it defines its very meaning. It has conceptual as well as practical implications for the formulation of hydrodynamic theories and their application to the physics of quark-gluon plasma as well as other fields – a point to which we shall return at the end of this Letter.

The hydrodynamic attractor was first identified explicitly in the differential equations of hydrodynamics [10, 12]. An outstanding problem is the determination of such attractors at the microscopic level [3, 13]. The most promising context for such a calculation is $\mathcal{N} = 4$ SYM, where holography provides powerful computational methods. Results obtained using these meth-

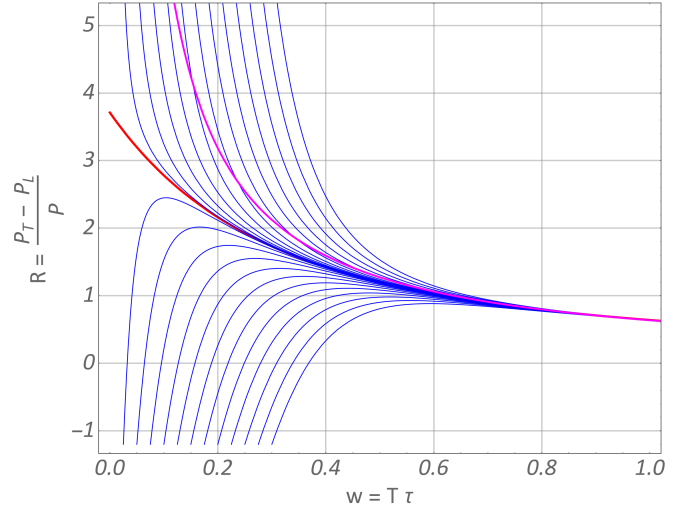


FIG. 1. Numerical solutions of the pressure anisotropy for Bjorken flow in second-order hydrodynamics (specifically, BRSSS [11]): the blue curves represent numerical solutions obtained for various initial conditions. They all decay to the numerically determined attractor solution [10], shown in red. The magenta curve represents the hydrodynamic gradient expansion truncated at first order in derivatives.

ods are the starting point for the approach pursued here: explicitly carrying out Borel summation of the gradient series calculated in Ref. [14]. We will show that the result indeed acts as an attractor for histories of Bjorken flow simulated using AdS/CFT.

A critical issue for Borel summation is the location of singularities of the analytic continuation of the Borel transform. These singularities reflect the spectrum of nonhydrodynamic modes – both at the microscopic level [14] and in hydrodynamics [10, 12]. An important testing ground for the feasibility and robustness of Borel summation of the gradient series of $\mathcal{N} = 4$ SYM is the hydrodynamic theory proposed in [15], which we will refer to as HJSW. This theory, unlike BRSSS hydrodynamics [11], extends Navier-Stokes theory by adding de-

degrees of freedom which mimic the leading (least-damped) nonhydrodynamic modes of $\mathcal{N} = 4$ SYM plasma, which translates to the same leading singularities [12] as those identified in Ref. [14].

An important point here is that the hydrodynamic gradient series is the leading element of a transseries [10], and in general the higher order elements (“instanton sectors”) play an important role in defining the summation properly. However, the instanton sectors are exponentially suppressed and it is tempting to ignore them as a first approximation. In the case of BRSSS theory this cannot be done, because the Borel transform has a singularity on the real axis, which leads to a complex ambiguity in the naive summation. This issue does not arise in $\mathcal{N} = 4$ SYM, nor in HJSW hydrodynamics, because singularities of the analytic continuation of the Borel transform occur off the real axis. Furthermore, including higher transseries sectors would introduce integration constants, which would have to be set appropriately to maintain accuracy of the summation at very early times. For practical purposes however, such accuracy may not be indispensable; in this paper we only consider summation of the “perturbative” part of the transseries – the original hydrodynamic gradient expansion.

As a confidence-building exercise we first perform the Borel summation of the gradient expansion in the case of HJSW hydrodynamics, where it is easy to check the validity of the answer. The result is unique, unambiguous, and coincides amazingly well with the attractor determined directly from the hydrodynamic equations. This sets the stage for the main theme of this Letter: the Borel summation of the gradient series of $\mathcal{N} = 4$ SYM. This is technically no more challenging than the calculation for HJSW theory, but its significance is that it provides an example of a hydrodynamic attractor obtained directly from a microscopic calculation. This result can only be fully appreciated by inspecting the behaviour of actual histories of boost-invariant expansion in $\mathcal{N} = 4$ SYM, which we will do later on. A very important point to note is that the attractor coincides with first order hydrodynamics at late times, but turns out to be quite distinct from it even at moderate times. This has implications of foundational nature for relativistic hydrodynamics. A fuller discussion of this result and its ramifications can be found in the concluding section.

Bjorken flow.– Throughout this paper we work with Bjorken flow [16], which imposes powerful simplifying symmetry constraints. We use proper time – rapidity coordinates τ, Y related to Minkowski lab-frame coordinates t, z by $t = \tau \cosh Y$ and $z = \tau \sinh Y$ where z is aligned along the collision axis. A system undergoing Bjorken flow has eigenvalues of the expectation value of the energy momentum tensor

$$T^\mu_\nu = \text{diag}(\mathcal{E}, \mathcal{P}_L, \mathcal{P}_T, \mathcal{P}_T)^\mu_\nu, \quad (1)$$

which are functions of the proper time τ alone. In a con-

formal theory, the conditions of tracelessness and conservation can be expressed as [17]

$$\mathcal{P}_L = -\mathcal{E} - \tau \dot{\mathcal{E}}, \quad \mathcal{P}_T = \mathcal{E} + \frac{1}{2} \tau \dot{\mathcal{E}}. \quad (2)$$

The departure of these quantities from the equilibrium pressure at the same energy density, $\mathcal{P} \equiv \frac{\mathcal{E}}{3}$, is a measure of how far a given state is from local equilibrium.

We will study the approach to equilibrium by examining the behavior of the pressure anisotropy

$$R \equiv \frac{\mathcal{P}_T - \mathcal{P}_L}{\mathcal{P}} \quad (3)$$

as a function of the dimensionless variable $w \equiv T\tau$, which plays the role of time. It is critically important to compare states of the system at different values of this dimensionless variable if we wish to see the attractor behaviour which is of central interest here.

The hydrodynamic attractor in hydrodynamics.– Hydrodynamic theories are described by sets of nonlinear partial differential equations. The key simplification brought by the assumption of Bjorken flow is that the equations of hydrodynamics reduce to ordinary differential equations. For example, the evolution equation for the pressure anisotropy in conformal BRSSS theory reads [3, 10]

$$C_{\tau_\pi} \left(1 + \frac{R}{12}\right) R' + \left(\frac{C_{\tau_\pi}}{3w} + \frac{C_{\lambda_1}}{8C_\eta}\right) R^2 = \frac{3}{2} \left(\frac{8C_\eta}{w} - R\right) \quad (4)$$

where the prime denotes a derivative with respect to w , and the dimensionless constants $C_\eta, C_{\tau_\pi}, C_{\lambda_1}$ are transport coefficients (whose values in the case of $\mathcal{N} = 4$ SYM are known, see e.g. Ref. [3]). This equation is nonlinear, but the attractor seen in Fig. 1 can be determined numerically by setting initial conditions appropriately [10]. It is important to observe that the attractor becomes indistinguishable from the first order truncation of the gradient series only for $w > 0.7$. For smaller values of w , the numerical solutions clearly decay to the attractor, not the truncated gradient series.

The pressure anisotropy in HJSW theory satisfies a second order nonlinear ordinary differential equation, whose exact form can be found in [3, 12], and a similar analysis leads to the numerical determination of its attractor solution (to which we shall return shortly). The point we wish to make at this juncture is that we cannot proceed in the same way in $\mathcal{N} = 4$ SYM, because there we cannot write down a closed differential equation like Eq. (4). To find the attractor in this case one has to find another way. The approach explored in this Letter is to sum the hydrodynamic gradient expansion, whose leading 240 coefficients were obtained using the AdS/CFT correspondence in Ref. [14]. In the following we discuss the properties of the series and the summation, using HJSW theory as a testing ground.

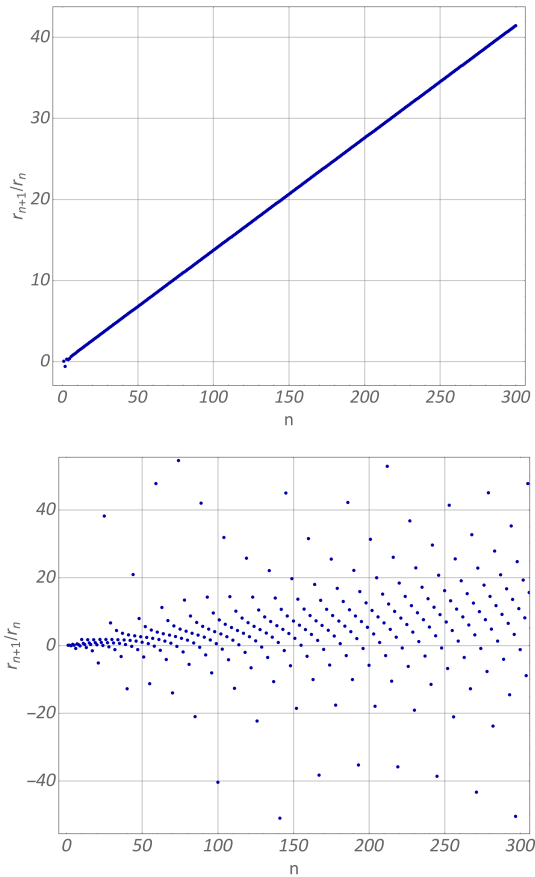


FIG. 2. The large order behaviour of the hydrodynamic series for BRSSS (top) and HJSW (bottom) hydrodynamics.

Large order behavior.— The general form of the gradient expansion for Bjorken flow in any conformal theory is

$$R(w) = \sum_{n=1}^{\infty} r_n w^{-n}. \quad (5)$$

The coefficients r_n have been calculated to high order in $\mathcal{N} = 4$ SYM [14], in kinetic theory [3, 7], as well as in various hydrodynamic theories [10, 12]. It is now well established that this series has a vanishing radius of convergence. In many cases, at large order n these coefficients grow in a way consistent with the Lipatov form [18]

$$r_n \sim \frac{n!}{A^n}, \quad (6)$$

where A is a real parameter. This formula implies linear behaviour of the ratio of neighbouring coefficients $r_{n+1}/r_n \sim n/A$. For the case of the gradient series in BRSSS hydrodynamics, this linear behaviour can be seen in the upper plot of Fig. 2. In the case of HJSW theory however the pattern is much more complex, as seen in the lower plot. The reason for this is that HJSW hydrodynamics instead of a single, purely damped nonhydrodynamic mode has a pair of modes with complex conju-

gate frequencies [12] (see also the recent review [3]). Furthermore, this signals that the hydrodynamic gradient expansion in this case is an element of a two-parameter transseries [19]; in BRSSS hydrodynamics the transseries involved only one parameter. The analysis of Ref. [12] can be used to find the following approximate formula describing the leading large n behavior

$$r_n \sim \frac{n!}{A^n} \cos \left((n + \beta_R)\phi - \psi + \beta_I \log \left(\frac{A}{n + \beta_R} \right) \right) \quad (7)$$

for some real numbers $A, \phi, \psi, \beta_R, \beta_I$ (the ψ appearing above is the phase of the Stokes constant of the transseries). Due to the oscillating factor this formula is not as useful as Eq. (6), but it does qualitatively capture the complex pattern in Fig. 2.

Importantly, if one plots the ratio of coefficients of the gradient expansion of $\mathcal{N} = 4$ SYM, calculated using the results in Ref. [14], one finds a picture very similar to the lower plot of Fig. 2. This happens because HJSW theory was constructed to reproduce the dominant non-hydrodynamic modes of $\mathcal{N} = 4$ SYM, which translates to the close similarity of the large-order behavior. This makes it a useful testbed for assessing the utility of Borel summation in this context, as explained below.

The attractor from Borel summation.— The Borel transform of the gradient series removes the dominant factorial growth of the expansion coefficients:

$$\mathcal{B}R(\xi) = \sum_{n=1}^{\infty} \frac{r_n}{n!} \xi^n. \quad (8)$$

This series will typically define an analytic function within a disc around the origin in the complex ξ plane. The Borel sum of the series is defined by the Laplace transform:

$$\mathcal{S}R(w) = w \int_{\mathcal{C}} d\xi e^{-w\xi} \tilde{\mathcal{B}}R(\xi), \quad (9)$$

where $\tilde{\mathcal{B}}R$ is the analytic continuation of the Borel transform Eq. (8) and $\mathcal{C}$ is a contour connecting 0 and ∞ .

The analytic continuation of $\mathcal{B}R(\xi)$ (performed using Padé approximants) necessarily contains singularities responsible for the vanishing radius of convergence of the original series. The singularities appearing in the cases of interest here have been discussed at length in the literature. For BRSSS theory one finds a branch point on the real axis [10], which introduces a complex ambiguity in the Borel summation, given by the difference in the values obtained for Eq. (9) by integrating above and below the cut. As stressed earlier, this complication does not arise in the case of $\mathcal{N} = 4$ SYM [14], or HJSW hydrodynamics [12] where the branch points appear away from the real axis. This means that one can perform the integral in Eq. (9) by integrating over real values of ξ from zero to infinity. In practice, this integral has to be performed numerically for a set of values of w .

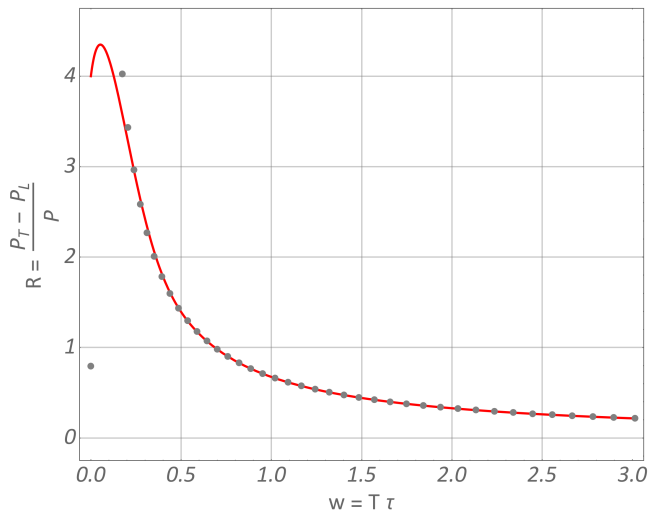


FIG. 3. The attractor in HJSW theory, calculated numerically from the hydrodynamic equations (red curve), compared with the results of Borel summation (gray dots).

To gauge the effectiveness of this method we begin with HJSW theory, which from this perspective offers qualitatively the same kind of challenge as $\mathcal{N} = 4$ SYM. The series can be calculated numerically to essentially arbitrarily high order [12], but here we will use only the first 240 terms, to have a fair testing ground for the $\mathcal{N} = 4$ SYM case, where the cost of calculating the coefficients is much higher, and at this time only 240 are available [14]. As is clear from Fig. 3, the Borel summation reproduces the numerically determined attractor closely down to $w \approx 0.4$, and is still quite reasonable at $w \approx 0.3$. To improve the accuracy of the summation below this value one would have to include contributions from higher transseries sectors.

The case of $\mathcal{N} = 4$ SYM.— The procedure described above can readily be applied to the hydrodynamic gradient expansion of $\mathcal{N} = 4$ SYM using the results of Ref. [14], where the expansion of the energy density in powers of $\tau^{-2/3}$ was calculated to order 240. Using formulae Eq. (2) and Eq. (3), the coefficients made available there can be translated into coefficients of the pressure anisotropy Eq. (5). As mentioned earlier, their ratios qualitatively follow the pattern described by the approximate formula Eq. (7) (and seen in the lower plot in Fig. 2). These coefficients can be used to calculate the Borel transform and its analytic continuation (using a diagonal Padé approximant), which one can integrate numerically for a range of values of w . The resulting curve is reproduced quite accurately by the rational function

$$R_A = \frac{560w^2 + 971355w - 99876}{1531200w^2 - 215760w + 48720} \quad (10)$$

for essentially all values of $w > 0$.

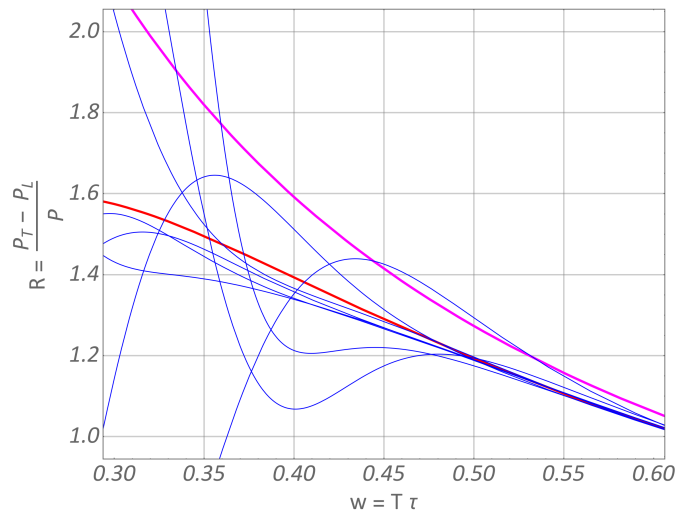


FIG. 4. The attractor of $\mathcal{N} = 4$ SYM plasma (the red curve), along with results of numerical simulations using AdS/CFT (blue curves); the magenta curve represents the hydrodynamic gradient expansion truncated at first order.

The attractor determined by Borel summation as described above can be expected to be reliable down to $w \approx 0.3 - 0.4$, but its utility is best judged by comparing it with the results of numerical holography simulations of Bjorken flow. A large number of such flow histories was studied in Ref. [20] (following the earlier work of Refs. [5, 21]). These solutions were found by randomly generating sets of consistent initial conditions for the bulk Einstein equations, constructing the spacetime numerically, and then using the holographic dictionary to read off the proper-time evolution of the energy density of $\mathcal{N} = 4$ SYM. Each such solution can then be used to calculate the pressure anisotropy as a function of the dimensionless variable w using equations (2) and (3).

As discussed in detail in Ref. [20], the transition to hydrodynamics takes place for a range of values of w centered around $w = 0.65$. The distinguished role played by the attractor will be most prominently visible by considering those solutions which reach the hydrodynamic regime early, at values of w for which the attractor and the truncated gradient expansion differ appreciably. A selection of such solutions is plotted in Fig. 4, along with an interpolation function fitted to the result of the Borel summation. We see there the same kind of striking behavior as displayed at the level of hydrodynamics in Fig. 1. Note that those histories which reach the attractor early do so while the pressure anisotropy significantly exceeds 100%, showing that this observable exhibits universal behaviour even in the highly nonequilibrium regime. Note also that the attractor is distinguishable from the truncated gradient series at a level exceeding 10% for values of $w \approx 0.45$.

The calculation reported here leaves a number of open problems. One of them is the explicit computation of leading transseries coefficients in the lowest instanton sectors. Such a calculation would make it possible to extend the range where the summation can be trusted to lower values of w . It would also allow us to verify that the resurgence relations written down in [12] connecting coefficients of different transseries sectors are satisfied. Another interesting problem to pursue would be a calculation of the $\mathcal{N} = 4$ SYM attractor directly in the holographic representation. An attempt of this kind was recently made by Romatschke [13], who used the idea of a “slow-roll” approximation to try to determine the special set of initial conditions corresponding to the attractor solution, which he was then able to estimate by evolution using the bulk Einstein equations. The results presented in that paper appear to be qualitatively consistent with the findings reported here.

Conclusions.— The goal of hydrodynamics is to mimic universal, late-time behaviour of systems tending toward equilibrium [3]. The BRSSS philosophy [11], which can be seen as an incarnation of the effective field theory paradigm, tells us to match leading terms of the gradient expansion of hydrodynamics with the corresponding terms calculated in the underlying microscopic theory. The developments of the past couple of years suggest that it could make sense to set a more ambitious goal: to try to reproduce the attractor of the underlying theory at the level of hydrodynamics. While the attractor matches low orders of the gradient series at sufficiently late times, earlier on it is different, and the difference depends on the parameters of the theory. Even for $\mathcal{N} = 4$ SYM the attractor is quite distinct from first (or second order) hydrodynamics. For QCD plasma, which most probably has a larger relaxation time, this distinction should be even more pronounced. This could have important consequences for the interpretation of observables sensitive to early time dynamics.

The idea that higher orders of the gradient expansion may be relevant for real-world physics is in itself not new [22–24]. However, the task of matching the attractor requires very good control of the higher orders and some understanding of the nonhydrodynamic sector – both at the microscopic level and in hydrodynamics.

The emerging picture is very promising, but it is too early to say how generic it is. At this point it would be most useful and interesting to find tractable examples which relax some of the technical elements which we have relied on, such as boost invariance and conformal symmetry. An important point will be to understand which observables reveal attractor behavior. Any progress should be of interest not only in the context of quark-gluon plasma, but also for other areas of physics [25, 26].

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